

Lecture 6: Continuous Random Variables, PDFs, CDFs

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0.1 Why do we need a new kind of random variable?

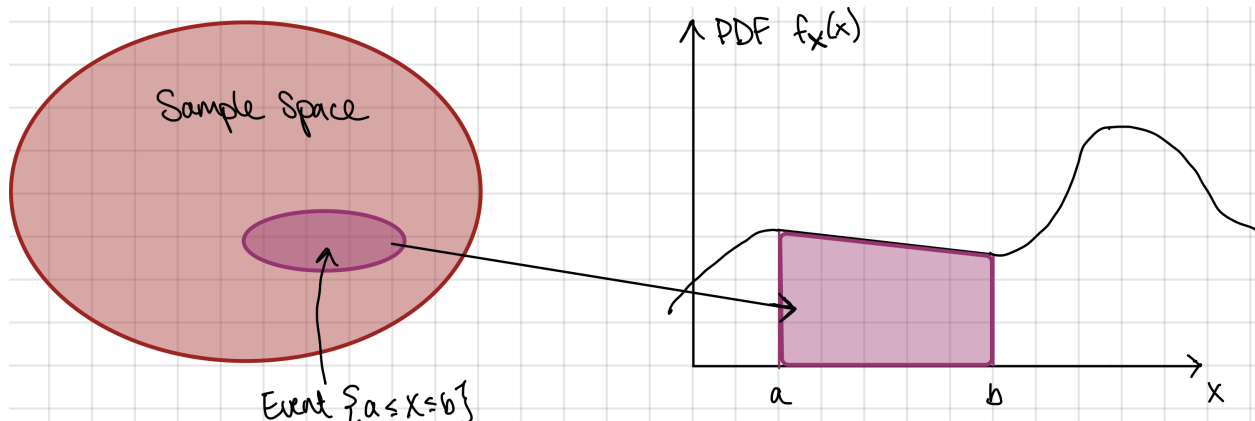
Every random variable so far has been **discrete**: a finite or countably infinite list of possible values, each with its own probability mass. But plenty of quantities we'd like to model — a waiting time, a position, a measurement error — can take *any* real value in an interval. There's no way to list them out one at a time.

i Definition: Continuous random variable, PDF

A random variable X is **continuous** if there is a function $f_X(x) \geq 0$, called its **probability density function (PDF)**, such that for every interval $[a, b]$,

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx, \quad \int_{-\infty}^{\infty} f_X(x) dx = 1.$$

Instead of bars of probability sitting on individual points (a PMF), probability is now **area under a curve**.



0.2 Example: the continuous uniform random variable

i Definition: Uniform(a, b)

$$f_X(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise.} \end{cases}$$

Every value in $[a, b]$ is “equally likely,” in the sense that the density is flat — probability of a sub-interval depends only on its *length*, not its location.

```
a_u, b_u = 2, 5

def f_uniform(x):
    x = np.asarray(x, dtype=float)
    return np.where((x >= a_u) & (x <= b_u), 1 / (b_u - a_u), 0.0)

xs = np.linspace(a_u - 1, b_u + 1, 400)
fig, ax = plt.subplots(figsize=(6, 3))
ax.plot(xs, f_uniform(xs), color='black')
ax.set_ylim(0, 1 / (b_u - a_u) * 1.3)
ax.set_xlabel('$x$')
ax.set_ylabel('$f_X(x)$')
ax.set_title(f'Uniform({a_u},{b_u}) density')
plt.tight_layout()
plt.show()

total, _ = quad(f_uniform, a_u - 1, b_u + 1)
p_3_4, _ = quad(f_uniform, 3, 4)
print(f"total area:          {total:.4f}")
print(f"P(3 <= X <= 4) =    {p_3_4:.4f}    (should be 1/3 = {1/3:.4f}, since [3,4] is 1/3 of [2,5])")

NameError: name 'np' is not defined

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NameError                                Traceback (most recent call last)
/var/folders/wm/5fzls3k12hx43mtc27zhd_mc0000gn/T/ipykernel_80416/1301219406.py in <module>
      5     return np.where((x >= a_u) & (x <= b_u), 1 / (b_u - a_u), 0.0)
      6
----> 7 xs = np.linspace(a_u - 1, b_u + 1, 400)
      8 fig, ax = plt.subplots(figsize=(6, 3))
      9 ax.plot(xs, f_uniform(xs), color='black')
NameError: name 'np' is not defined
```

0.3 Example: a PDF can take arbitrarily large values

⚠ A density is *not* a probability

Nothing in the definition requires $f_X(x) \leq 1$ — only that the *total area* equals 1. Squeeze the same unit of probability into a short enough interval and the height of the density can be as large as you like.

i Fact: $P(X = x_0) = 0$ for continuous X

For any single point x_0 ,

$$P(X = x_0) = \int_{x_0}^{x_0} f_X(x) dx = 0,$$

since we're integrating over an interval of zero width. Continuous random variables put *no* probability mass on individual points — only on intervals of positive length.

0.4 Expectation of a continuous random variable

i Definition: Expectation

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

The rules we've seen previously generalize once we replace summations with integrals!

i Theorem: LOTUS for continuous random variables

For any function g ,

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

i Definitions: moments and variance

The k -th **moment** of X is $E[X^k]$. **Variance** is defined exactly as in the discrete case,

$$\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2,$$

and the shortcut formula's proof used only linearity — nothing about summing versus integrating — so it carries over unchanged.

🔥 In-class exercise

Prove that for constants a, b and a continuous random variable X :

$$E[aX + b] = aE[X] + b \quad \text{and} \quad \text{Var}(aX + b) = a^2 \text{Var}(X).$$

Solution: identical to L4's discrete proof, with sums replaced by integrals.

$$E[aX + b] = \int (ax + b) f_X(x) dx = a \int x f_X(x) dx + b \int f_X(x) dx = aE[X] + b.$$

$$\text{Var}(aX + b) = E[(aX + b - aE[X] - b)^2] = E[a^2(X - E[X])^2] = a^2 \text{Var}(X).$$

Every fact about expectation and variance built on linearity in L4 carries over to continuous RVs with zero extra work.

0.5 The Exponential random variable

The continuous “cousin” of the Geometric — instead of counting *trials* until an event, it measures the *waiting time* until an event that can happen at any instant.

i Definition: Exponential(λ)

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0, \end{cases} \quad \lambda > 0.$$

$$E[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}$$

(both derivable by integration by parts) — the same shape of answer as the Geometric(p) mean $1/p$, with λ playing the role of a *continuous rate* in place of p .

Example: customers arrive at a coffee shop at an average rate of $\lambda = 3$ per hour, memorylessly (the time since the last arrival tells you nothing about when the next one comes). The waiting time X until the next customer is Exponential(3), measured in hours, so $E[X] = \frac{1}{3}$ hour = 20 minutes.

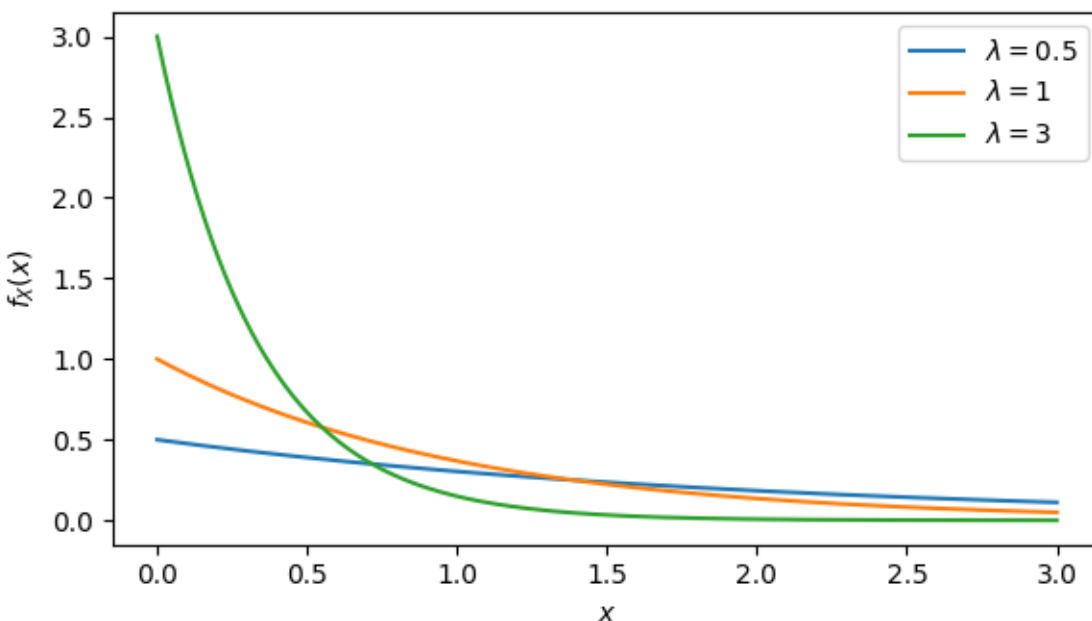
```
def f_exp(x, lam):
    x = np.asarray(x, dtype=float)
    return np.where(x >= 0, lam * np.exp(-lam * x), 0.0)

xs = np.linspace(0, 3, 400)
fig, ax = plt.subplots(figsize=(6, 3.5))
for lam in [0.5, 1, 3]:
    ax.plot(xs, f_exp(xs, lam), label=f'$\\lambda={lam}$')
ax.set_xlabel('$x$')
ax.set_ylabel('$f_X(x)$')
ax.legend()
plt.tight_layout()
plt.show()

lam = 3.0 # the coffee shop: 3 customers/hour
E_exp_num, _ = quad(lambda x: x * f_exp(x, lam), 0, 50)
E_exp2_num, _ = quad(lambda x: x ** 2 * f_exp(x, lam), 0, 50)
Var_exp_num = E_exp2_num - E_exp_num ** 2

rng = np.random.default_rng(151)
wait_times = rng.exponential(1 / lam, size=1_000_000) # numpy's scale parameter is 1/lambda

print(f"lambda = {lam}")
print(f"E[X]: formula 1/lambda = {1/lam:.4f} quad = {E_exp_num:.4f} simulated = {wait_times.mean():.4f}")
print(f"Var(X): formula 1/lambda^2 = {1/lam**2:.4f} quad = {Var_exp_num:.4f} simulated = {wait_times.var():.4f}")
```



```
lambda = 3.0
E[X]:  formula 1/lambda    = 0.3333    quad = 0.3333    simulated = 0.3327
Var(X): formula 1/lambda^2 = 0.1111    quad = 0.1111    simulated = 0.1112
```

0.6 Cumulative Distribution Functions

i Definition: CDF

For *any* random variable X — discrete or continuous —

$$F_X(x) = P(X \leq x).$$

Unlike the PMF or PDF, the CDF is defined the *same way* regardless of type. And each of PMF, PDF, or CDF — whichever applies to X — **unambiguously pins down the entire distribution**: from any one of them, you can recover the others.

0.7 Properties of the CDF

Every CDF, discrete or continuous, satisfies:

- **monotonically nondecreasing**: $x_1 \leq x_2 \implies F_X(x_1) \leq F_X(x_2)$,
- $F_X(x) \rightarrow 0$ as $x \rightarrow -\infty$, and $F_X(x) \rightarrow 1$ as $x \rightarrow \infty$,
- for **discrete** X : F_X is a **piecewise-constant step function** — flat except for a jump of size $p_X(x)$ at every x in the range of X ,
- for **continuous** X : F_X is **continuous** everywhere (no jumps at all — consistent with $P(X = x_0) = 0$ for every x_0).

Relation between PDF and CDF, for X continuous:

$$F_X(x) = \int_{-\infty}^x f_X(t) dt, \quad \text{equivalently} \quad f_X(x) = F'_X(x)$$

wherever F_X is differentiable — the Fundamental Theorem of Calculus!

```

from scipy.stats import binom

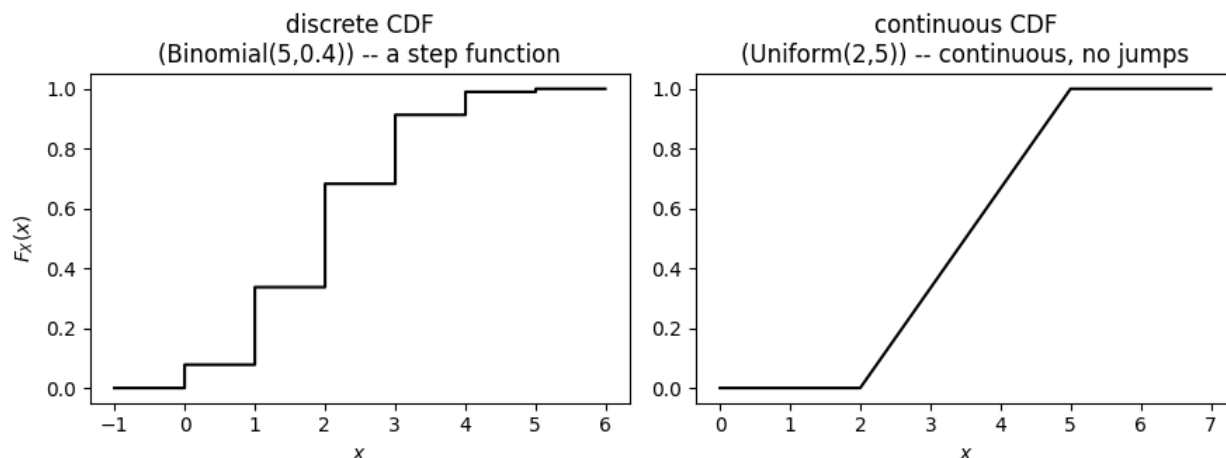
fig, axes = plt.subplots(1, 2, figsize=(9, 3.5))

# discrete: Binomial(5, 0.4) CDF -- a step function
n_b, p_b = 5, 0.4
xs_d = np.linspace(-1, 6, 700)
cdf_d = binom.cdf(np.floor(xs_d), n_b, p_b)
axes[0].step(xs_d, cdf_d, where='post', color='black')
axes[0].set_title('discrete CDF\n(Binomial(5,0.4)) -- a step function')
axes[0].set_xlabel('$x$')
axes[0].set_ylabel('$F_X(x)$')

# continuous: Uniform(2,5) CDF -- continuous, and (in this special case) piecewise linear
xs_c = np.linspace(0, 7, 400)
cdf_c = np.clip((xs_c - a_u) / (b_u - a_u), 0, 1)
axes[1].plot(xs_c, cdf_c, color='black')
axes[1].set_title('continuous CDF\n(Uniform(2,5)) -- continuous, no jumps')
axes[1].set_xlabel('$x$')

plt.tight_layout()
plt.show()

```



0.8 Geometric and Exponential CDFs

$$X \sim \text{Geometric}(p) : F_X(k) = 1 - (1 - p)^k, \quad k = 1, 2, \dots$$

$$X \sim \text{Exponential}(\lambda) : F_X(x) = 1 - e^{-\lambda x}, \quad x \geq 0.$$

Same shape of formula — because the Exponential really is the **continuous limit** of the Geometric. Chop a unit of time into n tiny slices, let each slice independently “succeed” with probability $p = \lambda/n$, and count *time* (not *trials*) until the first success. As $n \rightarrow \infty$, the Geometric CDF — rescaled so trial number k becomes time k/n — converges to the Exponential(λ) CDF.

💡 Aside

Both distributions are **memoryless** — the property we used to derive the Geometric’s mean carries over exactly to the Exponential, and we’ll put it to work again later this course.

```

lam = 2.0
t = np.linspace(0, 3, 400)
exp_cdf = 1 - np.exp(-lam * t)

fig, ax = plt.subplots(figsize=(6, 3.5))
ax.plot(t, exp_cdf, 'k--', lw=2, label=f'Exponential( $\lambda={lam}$ ) CDF')

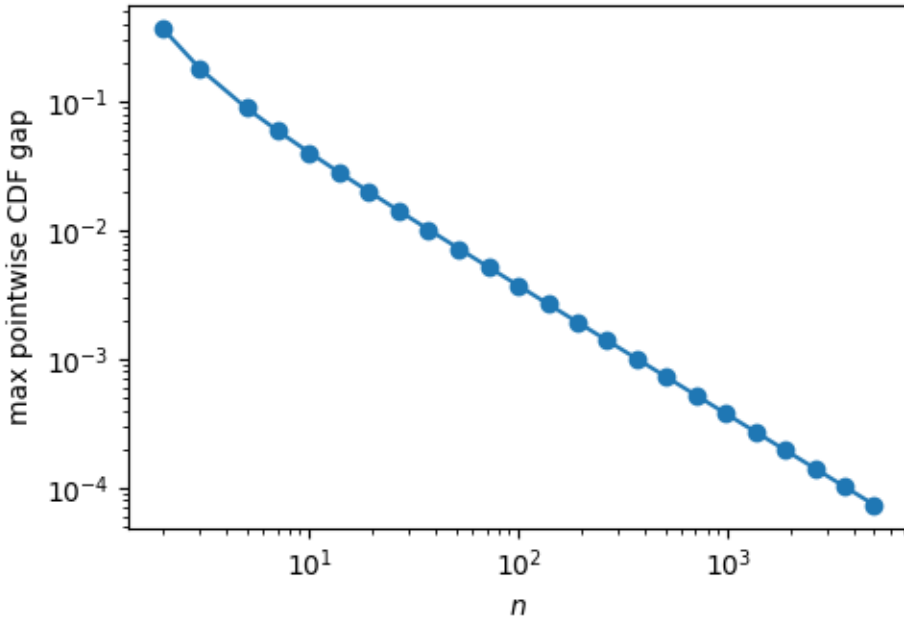
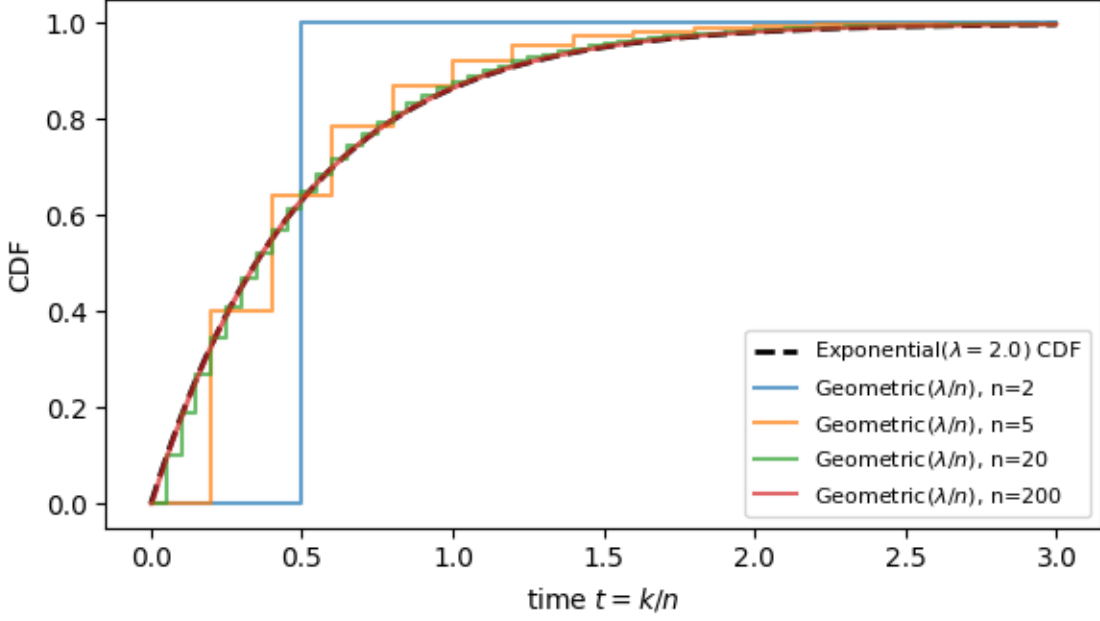
for n in [2, 5, 20, 200]:
    p = lam / n
    k_vals = np.arange(0, int(3 * n) + 1)
    geo_cdf = 1 - (1 - p) ** k_vals
    ax.step(k_vals / n, geo_cdf, where='post', alpha=0.7, label=f'Geometric( $\lambda/n$ ), n={n}')

ax.set_xlabel('time $t = k/n$')
ax.set_ylabel('CDF')
ax.legend(fontsize=8)
plt.tight_layout()
plt.show()

# max pointwise CDF gap, shrinking as n grows
n_range = np.unique(np.geomspace(2, 5000, 25).astype(int))
max_gaps = []
for n in n_range:
    p = lam / n
    k_vals = np.arange(0, int(5 * n) + 1)
    geo_cdf = 1 - (1 - p) ** k_vals
    t_vals = k_vals / n
    exp_at_t = 1 - np.exp(-lam * t_vals)
    max_gaps.append(np.max(np.abs(geo_cdf - exp_at_t)))

fig2, ax2 = plt.subplots(figsize=(5, 3))
ax2.loglog(n_range, max_gaps, 'o-')
ax2.set_xlabel('$n$')
ax2.set_ylabel('max pointwise CDF gap')
plt.tight_layout()
plt.show()

```



0.9 Example: continuous RV in randomized trace estimation

Recall the randomized trace estimator: for $A \in \mathbb{R}^{n \times n}$, draw $z \in \mathbb{R}^n$ with i.i.d. mean-0, variance-1, pairwise independent entries and set $\tau = z^\top A z$. The unbiasedness proof only ever used

$$E[z_i^2] = 1 \quad \text{and} \quad E[z_i z_j] = 0 \quad (i \neq j),$$

which followed from mean 0, variance 1, and pairwise independence — **nothing about z_i being discrete**. So a continuous z_i works exactly as well.

💡 A continuous alternative

Instead of Rademacher entries $z_i = \pm 1$, draw z_i i.i.d. from a continuous distribution with mean 0 and variance 1 — for concreteness, the (bell-shaped) standard Gaussian, whose PDF we'll meet formally in a later lecture. The same proof works exactly, just a different distribution!

```
rng = np.random.default_rng(151)
d = 50
A = rng.standard_normal((d, d)) + 5 * np.eye(d)
true_trace = np.trace(A)

num_samples = 20_000
Z_rad = rng.choice([-1.0, 1.0], size=(num_samples, d)) # discrete: Rademacher
Z_gauss = rng.standard_normal((num_samples, d))         # continuous: standard Gaussian

tau_rad = np.einsum('si,ij,sj->s', Z_rad, A, Z_rad)
tau_gauss = np.einsum('si,ij,sj->s', Z_gauss, A, Z_gauss)

print(f"true trace(A) = {true_trace:.4f}")
print(f"mean of tau, Rademacher z (discrete) = {tau_rad.mean():.4f}")
print(f"mean of tau, Gaussian z (continuous) = {tau_gauss.mean():.4f}")

true_trace(A) = 250.0783
mean of tau, Rademacher z (discrete) = 249.9342
mean of tau, Gaussian z (continuous) = 249.5031
```